



Probabilistic (Graphical) Models

Global Introduction

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'a'/'z' keys to browse the slides, 'o' for an overview, 'd' for dark/light mode

Objectives for the (P)(G)(M) Course

- Understanding
 - what is a probabilistic model
 - from fitting a distribution
 - to huge generative models
 - what is a Probabilistic Graphical Model (PGM)
 - what are the inference/learning methods
 - how this relates/differs/interacts with other approaches (e.g. deep learning, VAE, diffusion)
- Skills, be able to
 - model a problem with probabilities
 - reason using probabilities
 - read and write graphical models
 - select (and implement) models and inference methods

Presentation

Syllabus for the PGM Course / Disclaimer

- α. Introduction, uncertainty, generative models
- β. Probability, statistics, prob. modelling, maximum likelihood
- γ. Bayesian principles and bayesian learning
- δ. Bayesian Networks, Independence, Factorization, Sampling
- ε. Inference Methods for Probabilistic Models
- ζ. Variational Auto-Encoders (and variational inference)
- η. A Zoo of (advanced) Bayesian Networks
- θ. Deep Learning meets Probabilistic Models
- ι. Probabilistic Programming Languages (and Frameworks)
- κ. Student Project

Random Quotes

colleague: ... probabilistic models are just condition probabilities ...

student: ... in your probability course ...

colleague: ... never understood the bayesian stuff ...

Before we start

Survey

- at ease in Python?
- at ease with numpy?
- at ease with a deep learning framework?
- ... which one?
- already seen some graphical models?
- at ease with probabilities?
- already seen some probabilistic models?
- somewhat color blind?

Overview

- Presentation
- Logistics/resources
- Motivation/Introduction

Logistics/resources

Logistics

- 4 ECTS, 20h of courses $\approx 10 \times 2h$
- Final exam
- Quick exams possible at every lesson (on previous lessons and/or homework exercises)
- Project
- Rules of the game
 - be active, participate
 - ask questions
 - interrupt when needed
 - correct/report mistakes, incoherent/wrong notations and typos
- Propose subjects for later lessons
- Course reference <https://learn.heeere.com/2025-pgm-5f68/>
- Interesting resources
 - (PGM book by Daphne Koller (much detailed, very exhaustive) and Coursera)
 - V. Kuleshov and S. Ermon, very good: <https://ermongroup.github.io/cs228-notes/>
 - S. Ermon and A. Grover, oriented toward deep: <https://deepgenerativemodels.github.io/notes/>

More: interesting resources

- about notational shortcuts in probability <https://www.johndcook.com/blog/2008/02/23/everything-begins-with-p/>
- Some books
 - Probabilistic Graphical Models, Daphne Koller
 - Pattern Recognition and Machine Learning, Christopher M. Bishop
 - Machine Learning: A Probabilistic Perspective, Kevin Patrick Murphy
 - <https://probml.github.io/pml-book/> (pdf and all sources etc)
 - with draft of the new book
- Previous slide
 - <https://ermongroup.github.io/cs228-notes>
 - <https://deepgenerativemodels.github.io/>
- 3b1b videos on bayes rule
 - must watch explanation of bayes rule <https://www.youtube.com/watch?v=HZGCoVF3YvM>
 - trivial proof of bayes theorem https://www.youtube.com/watch?v=U_85TaXbeIo&t=0s
 - interactive area plot <https://www.skobelevs.ie/BayesTheorem/>
 - bayes factor <https://www.youtube.com/watch?v=IG4VkPoG3ko>
- Bayesian deep learning tutorial ICML2020 <https://www.youtube.com/watch?v=E1qhGw8QxqY> (Andrew Gordon Wilson)
- Historical talk on Bayesian Neural Nets at NIPS 2016, <https://www.youtube.com/watch?v=FD8l2vPU5FY> (Zoubin Ghahramani)

Motivation/Introduction

Decision Making Under Uncertainty

Spring School presentation as an introduction (s1-s16)

Supervised vs Unsupervised Learning

- Data? e.g. for supervised binary classification with images
- Goal?
- Formalizing learning? Supervised learning as optimization?
- Using a supervised-ly learned model?
- Generalization guarantees?
- Data for unsupervised learning?
- Goal?
- Formalizing learning? Optimization? Loss?
- Using a generative model?
- Generalization guarantees?
- Creativity?

Generative Models: the simplest

- Simplest data?
- Simplest model?
- Criteria?
- Optimal?

Generative Models: the next simplest

- Less simple data?
- Simplest model?
- Criteria?
- Optimal?

Generative Models: going further

- continuous data
- more complex models?

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Next

- Back to basics of probabilities + what it is to be "Bayesian"

For next time:

- Refresh your probability skills + re-derive a few max-likelihood estimators
- Watch bayes rule derivation https://www.youtube.com/watch?v=U_85TaXbeIo (3min45)
- Challenge:
 - A coin was flipped 8 times, you observed 3H (heads) and 5T (tails).
 - "If the next 5 flips are all T, I give you 1M€, else I take the money you bet"
 - How much do each of you bet on that? I'll take the highest bet.